

Fractions

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1 Addition

$$\frac{a}{b} + \frac{c}{b} = \frac{a+c}{b}$$

$$\frac{a}{b} + \frac{c}{d} = \frac{a \times d + b \times c}{b \times d}$$

2 Soustraction

$$\frac{a}{b} - \frac{c}{b} = \frac{a-c}{b}$$

$$\frac{a}{b} - \frac{c}{d} = \frac{a \times d - b \times c}{b \times d}$$

3 Multiplication

$$\frac{a}{b} \times \frac{c}{b} = \frac{a \times c}{b \times b} = \frac{a \times c}{b^2}$$

$$\frac{a}{b} \times \frac{c}{d} = \frac{a \times c}{b \times d}$$

4 Division

$$\frac{\frac{a}{b}}{\frac{c}{d}} = \frac{a}{b} \times \frac{d}{c}$$

EXERCICES

Exercice 1 : Calculer :

- $\frac{1}{4} \times \frac{1}{2}$
- $\frac{10}{20} - \frac{3}{12}$
- $\frac{3}{5} + \frac{2}{7}$
- $\frac{8}{9} - \frac{1}{6}$
- $\frac{4}{11} \times \frac{3}{8}$
- $\frac{7}{10} \div \frac{2}{5}$

Exercice 2 : Déterminer A :

$$A = \frac{1}{\frac{\frac{3}{2}}{\frac{1}{4}}}$$

Exercice 3 : Simplifie les expressions suivantes :

- $A = \frac{2^5 \times 2^3}{2^4}$
- $B = \frac{3^4 \times 3^2}{3^5}$
- $C = \frac{5^3}{5^2 \times 5}$
- $D = \frac{(2^3)^2}{2^4}$

Exercice 4 : Calculer la dérivée de f défini par $D_f = [0; +\infty[$:

$$f(x) = \frac{2\sqrt{x}(x+3)}{x+3}$$